

**Author(s):**

Fatma Alsaghir (220315105)

Jalil Guliyev (220315102)

Mahammadali Aliyev (220315090)

Heydar Mehdizade (220315091)

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| Ontology Requirements Specification Document |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1                                            | <b>Purpose</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                              | This ontology is designed to model and organize the main entities and relationships in an e-learning system, enabling structured representation of academic activities such as enrollment, teaching, submission, and evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2                                            | <b>Scope</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                              | <p>This ontology focuses on an E-learning system, including students, instructors, courses, and academic activities.</p> <p>Included:</p> <ul style="list-style-type: none"><li>• Students, Instructors, Courses, Modules, Assignments, Submissions, Grades</li><li>• Enrollment relationships and teaching relationships</li><li>• Assignment submission and evaluation</li></ul> <p>Excluded:</p> <ul style="list-style-type: none"><li>• Authentication systems (login, passwords)</li><li>• Technical platform details (video streaming, storage)</li><li>• Communication features (chat, forums)</li><li>• Advanced analytics or AI features</li></ul> |
| 3                                            | <b>Implementation Language (optional)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                              | The ontology is implemented in OWL/RDF and developed using Protégé.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4                                            | <b>Intended End-Users (optional)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                              | The intended users of this ontology are students, instructors, and academic administrators who use e-learning platforms for learning, teaching, and course management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5                                            | <b>Intended Uses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                              | This ontology is intended to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|   | <ul style="list-style-type: none"> <li>Organize academic knowledge within an e-learning system</li> <li>Support structured querying of data</li> <li>Assist in understanding relationships between students, courses, assignments, and instructors</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6 | <b>Ontology Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|   | <b>a. Non-Functional Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|   | <ul style="list-style-type: none"> <li><b>NFR1:</b> The ontology shall be implemented in OWL/RDF (developed using Protégé).</li> <li><b>NFR2:</b> The class hierarchy shall be clean, non-redundant, and logically structured.</li> <li><b>NFR3:</b> All object and data property domains and ranges shall reference defined classes only.</li> <li><b>NFR4:</b> The ontology shall be logically consistent and pass OWL reasoner validation.</li> <li><b>NFR5:</b> Enrollment, prerequisite, and deadline concepts shall be represented as properties, not classes.</li> <li><b>NFR6:</b> The ontology shall exclude authentication systems, communication features, and platform-specific technical details.</li> <li><b>NFR7:</b> The ontology should be extensible and reusable for future enhancements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |
|   | <b>b. Functional Requirements: Lists or tables of requirements written as Competency Questions and sentences</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|   | <p>Core Competency Questions</p> <ul style="list-style-type: none"> <li><b>CQ1:</b> Which students are enrolled in a specific course?</li> <li><b>CQ2:</b> Which instructor teaches a specific course?</li> <li><b>CQ3:</b> Which assignments belong to a specific course?</li> <li><b>CQ4:</b> Which students submitted a specific assignment?</li> <li><b>CQ5:</b> What grade did a student receive for an assignment?</li> <li><b>CQ6:</b> Which courses have prerequisites?</li> <li><b>CQ7:</b> Which assignments have a deadline?</li> <li><b>CQ8:</b> Which modules belong to a specific course?</li> <li><b>CQ9:</b> What is the submission date for a student's submission?</li> <li><b>CQ10:</b> Which courses does a specific instructor teach?</li> </ul> <p>Extended Competency Questions</p> <ul style="list-style-type: none"> <li><b>CQ11:</b> If Course A requires Course B, and Course B requires Course C, does Course A require Course C?</li> <li><b>CQ12:</b> Can a student be enrolled in multiple courses at the same time?</li> <li><b>CQ13:</b> Does each submission have exactly one grade?</li> <li><b>CQ14:</b> What is the average score across all submissions for a given course?</li> </ul> |

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|          | <ul style="list-style-type: none"> <li>• <b>CQ15:</b> Which students submitted an assignment after its deadline?</li> </ul>                                                                                                   |
| <b>7</b> | <b>Pre-Glossary of Terms (optional)</b>                                                                                                                                                                                       |
|          | <b>a. Terms from Competency Questions</b>                                                                                                                                                                                     |
|          | Student, Course, Instructor, Assignment, Submission, Module, Grade, Prerequisite, Deadline, Enrollment                                                                                                                        |
|          | <b>b. Terms from Answers</b>                                                                                                                                                                                                  |
|          | enrolledIn, teaches, hasModule, hasAssignment, submits, submittedBy, submissionFor, hasGrade, hasPrerequisite                                                                                                                 |
|          | <b>c. Objects</b>                                                                                                                                                                                                             |
|          | <ul style="list-style-type: none"> <li>• Person → Student, Instructor</li> <li>• AcademicContent → Course, Module</li> <li>• Assessment → Assignment<br/>Submission (separate class)</li> <li>• Evaluation → Grade</li> </ul> |